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Melanoma of the pleura mimicking malignant mesothelioma: a rare and rapidly progressive pleural disease

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A woman in her 60s presented to an emergency department in Western Australia with a week's history of dyspnoea and a new non-productive cough. She had a history of rheumatoid arthritis with a stable immunosuppressive regimen of baricitinib and methotrexate, as well as connective tissue disease-associated interstitial lung disease which had been stable for the past 5 years.

On examination, she was dyspnoeic and hypoxic requiring 4 L of oxygen via nasal prongs to maintain saturations greater than 92%. Other vital signs remained within normal limits. There was no palpable lymphadenopathy or hepatosplenomegaly. Cardiovascular examination was normal, and the calves were soft and non-tender without subcutaneous oedema. Breath sounds were reduced at the left lung base with diffuse fine inspiratory crackles throughout all lung fields. A blood panel revealed a mild thrombocytosis, and inflammatory markers were not significantly elevated. Biochemical analysis of the liver, thyroid and kidneys was unremarkable. Bedside thoracic ultrasound revealed a moderate-to-large, left-sided homogenous pleural effusion. A postcontrast CT thorax (figure 1A) confirmed the moderate left-sided effusion, with the additional finding of nodular circumferential pleural thickening inclusive of the mediastinal pleura. Given the patient's history of exposure to asbestos-related materials, the combination of epidemiological data and radiological findings strongly indicated a diagnosis of malignant mesothelioma.

A high-volume thoracentesis was performed on day 2 of admission with cytological analysis revealing abnormal epithelioid and plump spindle-shaped cells with highly pleomorphic nuclei. These abnormal cells were positive on

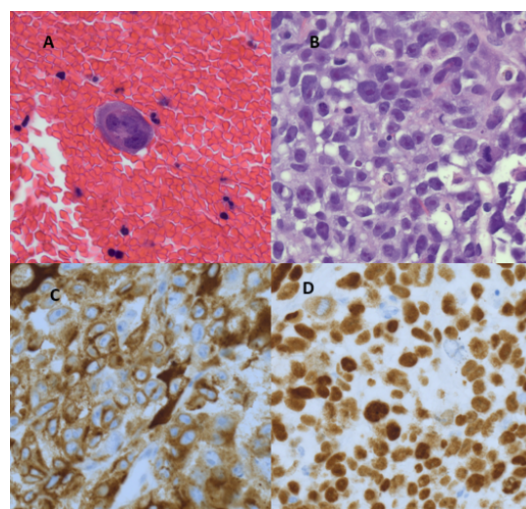


Figure 2 (A) Photomicrograph of melanoma cell in pleural fluid cytology. (B) Photomicrograph of melanoma cells in pleural biopsy. (C) Melan A immunohistochemical stain. (D) SOX-10 immunohistochemical stain.

immunohistochemical stains for SOX10, Melan A and S100, with negative stains for calretinin, EMA, p40 and WT1 (figure 2). A cytological diagnosis of malignant melanoma was made. Given the unusual appearance and location of the disease, a CT-guided pleural biopsy was obtained which confirmed abnormal cells of similar appearance with positive immunohistochemical stains for SOX10 and Melan A confirming the diagnosis.

Given the rarity of pleural melanoma and the absence of a personal history of cutaneous melanoma, a thorough dermatological and ocular examination was conducted. This assessment revealed no suspicious or concerning lesions. A positron emission tomography (PET) scan was performed revealing an intensely FDG (fluorodeoxyglucose)-avid left hemithorax with circumferential multi-nodular pleural thickening (figure 1B). No primary cutaneous or subcutaneous lesion was identified.

Rapid accumulation of the pleural fluid despite drainage required insertion of an indwelling pleural catheter with drainages of up to 1 L required every 48 hours for symptom relief. The disease was identified as BRAF and KIT wild type, making the use of targeted therapies unsuitable for this patient. Surgical intervention was ruled out due to the location of the tumour and the extensive disease burden. Given the patient's history of interstitial

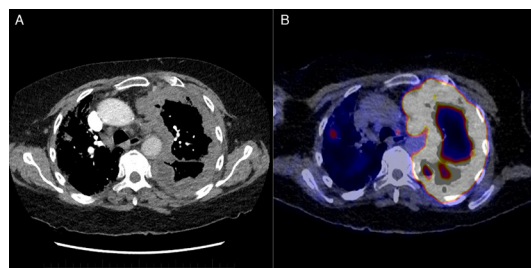


Figure 1 (A) Axial view CT of the chest demonstrating nodular circumferential pleural thickening. (B) Positron emission tomography revealing intensely FDG-avid left hemithorax. FDG, fluorodeoxyglucose.



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lung disease and the associated risk of immunotherapy-related pneumonitis, dual-agent immunotherapy with ipilimumab and nivolumab was deemed inappropriate. Instead, single-agent immunotherapy with nivolumab was initiated. Unfortunately, the disease progressed rapidly and showed no response to the treatment. The patient continued to deteriorate and transitioned to comfort-based care within 3 months of diagnosis.

DISCUSSION

Definition and epidemiology of malignant melanoma

Malignant melanoma is an aggressive cancer arising from melanocytes, the melanin-producing cells primarily located in the skin.¹ Although melanoma accounts for only 3% of all skin cancers, it is responsible for 60% of skin cancer-related mortality.² While it most commonly originates in the skin, melanoma can also develop in ocular or mucosal tissues.³ If not controlled early through surgical resection, metastases are frequent and can spread to the skin, subcutaneous tissues, skeleton, lymph nodes and visceral organs.

Investigation and management of malignant melanoma

The initial step in the evaluation of suspected melanoma involves a thorough clinical examination of the skin. Dermoscopy is a non-invasive technique that enhances the visualisation of skin lesions and has been shown to improve diagnostic accuracy compared with visual inspection.⁴

A definitive diagnosis of melanoma is established through biopsy with the most common type of biopsy being excisional, with removal of the entire lesion. Histopathological examination of the biopsy specimen is essential for confirming the diagnosis and determining the melanoma subtype, which can influence prognosis and treatment. Sentinel lymph node biopsy is performed in cases where the melanoma is thicker than 1 mm or has other high-risk features for staging purposes to guide further treatment decisions.⁵

Imaging studies play a critical role in the staging of melanoma, particularly when there is a suspicion of metastasis. Common modalities include CT or PET which are useful for evaluating regional lymph nodes and distant metastases, while MRI is beneficial for assessing brain metastases.

Genetic testing for mutations in genes such as BRAF, NRAS and KIT can provide insight into the tumour's biology and inform targeted therapy options, such as BRAF inhibitors, which have shown efficacy in treating advanced melanoma.⁶

The management of melanoma depends on the pathological and radiological staging of the tumour. Surgery, the cornerstone of treatment for localised melanoma, may also be an option for patients with oligometastatic disease, where metastases are limited in number and location.⁷

For metastatic melanoma, systemic therapies play a central role and are broadly categorised into immunotherapy and targeted therapy. Immune checkpoint inhibitors, such as PD-1 inhibitors (nivolumab), PD-L1 inhibitors (pembrolizumab), CTLA4 inhibitors (ipilimumab) or a combination of the above, are the primary treatment options and have shown remarkable efficacy in the setting of metastatic disease.⁸ For patients with BRAF V600 mutations, targeted therapy with BRAF and MEK inhibitors also offers significant improvements in both progression-free and overall survival.⁹

Melanoma of unknown primary

In approximately 3% of cases, melanoma is discovered in the lymph nodes or visceral organs without a detectable primary

site.¹⁰ Various hypotheses have been proposed to explain the origin of melanoma with an unknown primary, including the spontaneous regression of the primary tumour and the presence of ectopic melanocytes in lymph nodes or visceral organs. The distinction between primary melanoma of visceral organs and regressed cutaneous melanoma with visceral metastases remains a topic of debate, as reflected in the literature.¹¹

Pleural melanoma of unknown primary

We present the case of a woman in her 60s with pleural melanoma, presenting as a malignant effusion without an identifiable primary site. Initial differential diagnoses based on imaging findings included, but were not limited to, malignant mesothelioma, pleural metastases or carcinomatosis, lymphoma and primary lung cancer with pleural invasion. Melanoma of the pleura with unknown primary is exceedingly rare. To the best of our knowledge, only nine other cases have been reported in the English literature.^{12–20} Some of these cases were identified postmortem before the advent of modern diagnostic techniques which may have indeed found a primary site. Similar to prior cases, our patient exhibited a particularly aggressive form of the disease, which did not respond to conventional melanoma therapies.

Recommendations

Melanoma should be included in the differential diagnosis of visceral organ or nodal malignancies, even in the absence of a history of cutaneous, ocular or mucosal melanoma. With metastatic pleural effusions, a comprehensive analysis of pleural fluid is essential and should encompass cytological and immunohistochemical evaluations to confirm or disprove the diagnosis of melanoma. Given the aggressive nature of melanoma of unknown primary, prompt initiation of treatment is critical to improve patient outcomes.

Learning points

- ▶ Melanoma should be included in the differential diagnosis of visceral organ or nodal malignancies, even in the absence of a history of cutaneous, ocular or mucosal melanoma.
- ▶ With metastatic pleural effusions, a comprehensive analysis of pleural fluid is essential and should encompass cytological and immunohistochemical evaluations to confirm or disprove the diagnosis of melanoma.
- ▶ Given the aggressive nature of melanoma of unknown primary, prompt initiation of treatment is critical to improve patient outcomes.

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Case reports provide a valuable learning resource for the scientific community and can indicate areas of interest for future research. They should not be used in isolation to guide treatment choices or public health policy.

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